IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 11 November 2024, accepted 17 January 2025, date of publication 29 January 2025, date of current version 4 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3536205

RESEARCH ARTICLE

Striking the Balance: Evaluating Content Quality and Reward Dynamics in Blockchain Online Social Media

DAVIDE MANCINO[1], BARBARA GUIDI[2], ANDREA MICHIENZI[2], AND MARCO VIVIANI[1]
[1]Dipartimento di Informatica, Sistemistica e Comunicazione, Università degli Studi di Milano-Bicocca, 20127 Milan, Italy
[2]Dipartimento di Informatica, Università degli Studi di Pisa, 56127 Pisa, Italy
Corresponding author: Barbara Guidi (guidi@di.unipi.it)

ABSTRACT Various forms of decentralized solutions for social media development have been proposed in the last years to address some of the open problems of centralized ones, which include security issues, micro-targeting, opinion polarization, and various forms of information pollution. Regarding the latter issue in particular, some decentralized social media have implemented reward systems based on the blockchain technology with the stated purpose of incentivizing users to generate quality content, which can be seen as a simple mechanism of bottom-up content moderation in decentralized environments. However, users can still cheat the logic of such reward systems, to maximize their economic gain, regardless of content quality. In this paper, we focus on this problem by studying the actual relationship between the quality of User-Generated Content disseminated in the context of blockchain-based decentralized social media platforms, and the rewards provided to users by their reward systems. To this end, we analyze distinct sets of features characterizing quality content in social media, in association with distinct classifiers to identify content actually worthy of reward. By exploiting Steemit, one of the most well-known decentralized social media platforms based on blockchain, we construct a labeled dataset to both train and evaluate distinct configurations of the different feature sets in association with the considered classifiers. The performed evaluation shows the effectiveness of some of these configurations in identifying reward-worthy content, demonstrates that current reward systems are indeed unable through their mechanisms to promote actual quality content, and suggests the need for the development of further reward strategies allowing for better reward distribution and, ultimately, help pursue the original purpose of reward systems.

INDEX TERMS Blockchain online social media, content quality, decentralized online social networks, natural language processing, reward systems, social media, user-generated content.

I. INTRODUCTION

Today, *Online Social Networks and Media* (OSNEM) platforms represent one of the most popular Internet services, with approximately 60% of the World's population being active members.[1] People use OSNEMs for the most disparate reasons, including keeping in touch with friends and family, seeking distinct kinds of advice, being informed of news and events in a timely manner, advertising products, etc. Despite their positive aspects, OSNEMs are also afflicted by a number of issues, including dependence on a provider, censorship, violation of privacy, poor redistribution of value, opinion polarization, dissemination of misinformation, etc. [1], [2]. In this scenario, researchers and developers identified the centralized nature of these platforms as a possible aspect to change in order to provide new social services that are able to address the aforementioned issues, paving the way for *Decentralized Online Social Networks* (DOSNs). Decentralization can take many forms. *Federation*

The associate editor coordinating the review of this manuscript and approving it for publication was Xiaodong Xu.

[1]https://wearesocial.com/uk/blog/2022/01/digital-2022

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

is one of the most explored directions, as shown by the numerous independent decentralized platforms that compose the so-called *Fediverse* (e.g., Diaspora,[2] Mastodon, Pleroma, Pixelfed, and others).[3]

In recent years, blockchain technology is being considered for the decentralization of OSNEMs, giving birth to the so-called *Blockchain Online Social Media* (BOSM) platforms [3]. Thanks to blockchain technology, BOSM platforms can introduce *reward systems* to overcome various problems, mainly related to *content quality*. Indeed, users can obtain an economic reward based on their actual social contribution to the platform [3]. For example, in the Steem bluepaper,[4] it is stated that:

> *[...] This makes Steem a public publishing platform from which any Internet application may pull and share data while* ***rewarding*** *those who contribute the most* ***valuable content****.*

Such systems are usually based on the concept of *token economy* [4], and grant *cryptocurrency* to users that (should) create impactful or informative *User-Generated Content* (UGC), which other users can find interesting and worth read, like reports, tutorials, (genuine) breaking news, original artistic content, etc. Therefore, through the operation of such reward systems, the problem of *content moderation* in BOSMs could also be in fact addressed [5], in a bottom-up and decentralized way [6].

During the last few years, several BOSM platforms such as *Steemit* [7], *hive.blog*,[5] *Minds*,[6] *DeSo*,[7] *Yup* [8] and many others have introduced distinct decentralized reward systems; each of them is based on distinct characteristics. As an example, Steemit and hive.blog reward systems adopt a so-called *Proof-of-Brain* approach, where users of the social platform can evaluate the posts they see through an *upvote* if they think that the post deserves more reward, or a *downvote* in the opposite case. One week after the creation of each post, the final reward is computed and granted to the creator. Yup adopts an analogous approach, however, the one-week time window is removed, and the voting system is more complex, entailing five types of votes and the fact that each post can be voted in three categories.

In spite of this, reward systems in BOSMs can be partially cheated once the logic behind them is discovered. For instance, *bot* accounts, which are capable of performing automatic actions, including creating and publishing content, can be employed to maximize one's gain [9]. On top of that, users can team up and exchange favors to increase the visibility of each other posts or the output of the reward system [10]. In addition, the introduction of an economic facet can have serious repercussions on the social activity of users; these repercussions can even lead to phenomena that are completely contrary to the initial purpose of the BOSM [3], including the possible generation or increase of *information bubbles* [11] and/or *polarization* [12].

Based on the premises just mentioned, in this paper we aim to study the actual relationship between the quality of content generated on BOSMs and the rewards received, by means of content analysis. Currently, there are no works in the literature that attempt to study this relationship in detail. For this reason, we focus on several characteristics of quality (mostly) related to textual content, by considering distinct sets of features characterizing the posts collected within BOSMs. Such feature sets include stylistic features, readability features, grammar features, opinion-based features, metadata features, and textual representation features. Different classifiers are employed together with distinct feature set configurations to determine when a post should actually be reward-worthy based on its quality. To do this, we trained such classifiers on data gathered from one of the most important BOSMs, i.e., Steemit, and labeled by human assessors that we recruited for this study. Labeling has been performed against four possible categories, i.e., *Deserving of reward*, *Partially deserving of reward*, *Not deserving of reward*, and *Unable to evaluate*. The trained supervised models considered rely on the usage of *Support Vector Machines*, *Random Forests*, *Logistic Regression*, *Multilayer Perceptron*, and *XGBoost*.

The comparative experimental evaluation performed in this study presents promising results, under different perspectives: (*i*) it demonstrates the effectiveness of accurately identifying posts deserving of rewards by utilizing the identified quality features and the considered supervised approaches; (*ii*) it highlights that current reward systems are indeed unable through their mechanisms to promote actual quality content and an effective bottom-up mechanism for content moderation; and (*iii*) it suggests that BOSMs platforms, such as Steemit, could benefit from considering content analysis in their further developments, not only to achieve the objectives for which they introduced reward systems, but also to gain potential economic savings. The overall contributions of this work are, therefore, summarized as follows.

- We provide a comprehensive *analysis of the literature* concerning the assessment of *content quality* in social media and how this research theme have been studied and tackled in the context of decentralized solutions such as DOSNs and BOSMs, and related open issues;
- We perform *content quality analyses* on BOSM data by identifying and describing several *quality features* related to the content of posts generated, and their use in association with machine learning techniques to classify content concerning its quality and thus its being truly worthy of reward;
- We provide a *comparative evaluation* of the proposed machine learning solutions and related feature sets in order to evaluate the greater effectiveness of certain

[2] https://diasporafoundation.org/
[3] https://fediverse.party/
[4] https://steem.com/steem-bluepaper.pdf
[5] https://hive.blog/
[6] https://www.minds.com/
[7] https://www.deso.com/

configurations over others, specifically to assess the *importance* of the various quality features considered;
- We make *publicly available* a *custom-built dataset* designed to explore the considered problem (both for training and evaluation perspectives) through the involvement of human assessors who rated individual posts from a specific BOSM, namely Steemit, against four different categories of reward deservingness;
- We provide further *reward allocation analyses* to investigate the correlation between votes, quality, and rewards, possibly leading to patterns of manipulation that can compromise the fairness of reward allocation;
- Overall, we demonstrate that current reward strategies fail to effectively identify and appropriately reward valuable content. As a result, social platforms squander significant funds on posts that should not merit recognition, leading to an increase in the proliferation of bots and various forms of low-quality content [9], which is in direct contradiction to the original purpose for which these reward systems were developed.

The rest of this article is organized as follows. Section II provides an overview of the background concepts pertinent to our contribution, along with an examination of the current state-of-the-art work. Section III describes in detail the study carried out in this article, the quality feature sets identified, and their use in association with machine learning classifiers. Section IV illustrates the evaluation we performed on the feature set-classifier configurations illustrated in the previous section, on the dataset generated upon Steemit data, together with further analyses related to reward allocation. Finally, Section V concludes the paper, summarizing the outcomes of the work and providing possible future research directions.

II. BACKGROUND AND RELATED WORK

In this section, we introduce the background concepts of blockchain, reward system, and the way in which they are employed in BOSMS. In particular, we compare current BOSM platforms to show in detail the characteristics of their reward strategies. Finally, we propose an overview of content quality analysis in social media, both centralized and decentralized, and the importance of content moderation in such platforms, with respect to both types of architecture.

A. THE BLOCKCHAIN TECHNOLOGY

A *blockchain* is a chain of cryptographically linked blocks of transactions and represents one of the possible implementations of the *Distributed Ledger Technology* (DLT) [13]. Developers and researchers began considering the blockchain with the launch of Bitcoin [14] in 2008. In its simplest form, a blockchain supports one *transaction* type, that represents a transfer of its crypto asset. One of the most impactful innovations on the blockchain is the introduction of *smart contracts* [15] in *Ethereum* [16], which are pieces of code (the contracts) deployed on the blockchain with which users can interact. The execution of the contract code is triggered by a transaction and is performed by the *miners*, i.e., nodes that validate the transactions. Thanks to the introduction of smart contracts, complex applications can be implemented atop the blockchain [17], [18], such as decentralized exchanges, reward systems, and social media, just to name a few. As concerns this latter scenario, a specific kind of blockchain technology has been proposed, which is principally focused on social media platforms and proposes a rich set of transaction types that can model the users' social activity on them. One such example is the blockchain of Steem that, rather than storing transactions concerning cryptocurrency only, is able to store pieces of information like *user A created a post*, or *user B likes a post,* by using special transactions.

B. REWARD SYSTEMS

A *reward system*, in general, is designed to encourage people to participate in an event, activities, spending money, etc., in return for a reward [19]. In terms of blockchain-based reward systems for social applications, there are some important proposals, detailed here below.

Steem is the first blockchain proposed to support social applications, and one of its main innovations is the introduction of a *blockchain-based reward system* that takes into account the social activity of its users. Users create posts that can include several types of media, such as text, images, pictures, animated gifs, and videos. Other users can *vote* on posts, even their own, and votes are *weighted* by the stake of the user on the platform. If a post receives enough votes within one week of its creation, a reward is assigned to it. At least half of the reward is granted to the post creator, while the remaining is divided among its *curators* (i.e., users who voted on the post), where the first voters get a better reward. The main motivation behind introducing this reward system is to encourage creators to produce *high-quality content* for the platform, and curators to carefully evaluate the pieces of content produced by creators, following a *Proof-of-Brain* approach.[8] A similar approach is adopted by *Hive*, a blockchain born in March 2020 from a hard fork of the Steem blockchain. The blockchain was launched because of a hostile takeover that happened a few months prior to its launch. While most of the functionalities remained the same, Hive introduced some important changes concerning how users' stake is considered in most voting activities, including the computation of rewards.

Minds is a BOSM with a *hybrid* architecture. The social platform is available through a *centralized* set of servers, while rewards are granted in the form of an ERC-20-compliant token on the *Ethereum* blockchain.[9] The reward system considers a wide range of social activities rewardable, including posting comments, attracting new subscribers, and others. Each action deserving a reward contributes to a point system to evaluate the social attention generated by each user.

[8]https://steem.com/steem-bluepaper.pdf
[9]https://ethereum.org/

At the end of each day, the newly created tokens are awarded proportionally, based on the social attention generated.

Odysee,[10] is a video-sharing Website that uses the LBRY protocol to build a *decentralized* platform. LBRY is principally used to empower decentralized platforms in video sharing and social media applications. Odysee should constitute an alternative to YouTube, where users can upload videos and receive LBRY credits which are LBRY's own cryptocurrency that can be used to promote videos. In Odysee, several rewards can be obtained: a user can collect a *view master* reward, given based on the total number of videos watched since joining Odysee; a *coveted creator* reward, given based on the total number of verified followers (followers with verified accounts); a reward for *following creators*; a *weekly watch* reward, given for watching videos for at least three days in a given week; finally, it is possible to collect a reward for *reposting content* and for *inviting other users* onto Odysee.

Yup [8] is a *one-of-a-kind* approach because it is a reward system that works in parallel to any social application. Users can be active in their favorite OSNEM, like Facebook or Twitter, while expressing their *feedback* (on a 5-values scale) on Yup. If multiple users share the same feedback, a certain amount of reward is assigned to the content creator and all other users who expressed the same feedback. Users can vote on all kinds of content, including social media posts, and everything that can be uniquely identified by an URL. This makes Yup a platform for evaluating the Web and letting contributors earn rewards for their feedback.

In 2019, *Reddit* launched the *Reddit Community Points* (RCPs) [20], as a way to measure the community contribution of a user in a specific subreddit. Whenever a user creates a post for the `/r/EthTrader` subreddit, the system keeps track of the number of upvotes obtained by the post. Every four weeks, a special post by the moderators of the subreddit announces how tokens will be distributed. The community tokens are implemented as Ethereum's ERC-20 tokens.

DeSo, the *Decentralised Social blockchain*, is an innovative blockchain, similar to the one proposed by Steem, because it also stores the social activity of its users in specialized transactions. DeSo has a highly speculative nature, indeed users can *bet* on the social success of a content creator by buying the so-called *creator coins*. These coins serve the purpose of funding content creators and can be used to redistribute with the funders the wealth generated by the content creators.

Pixie is another social platform, based on photo sharing that encourages social activity with crypto rewards [21]. Additionally, the platform integrates the possibility of transforming one's posts into NFTs.

Table 1 shows a multiple-aspect comparison of the aforementioned blockchain-based reward systems. We consider, in particular, six specific characteristics that identify the most important social activities included in the reward systems:

1) *Content curation* shows if the reward system evaluates valuable content through *likes*, *dislikes*, and *reactions* in general;
2) *Content creation* highlights if the system rewards the creation of *social content*, such as a post;
3) *Other social activities* shows if other social activities are included in the reward strategy, such as *re-post* or *following* activities;
4) *Other monetization* considers if *non-social activities* are included in the reward system;
5) *Content quality* shows if the system takes into account some form of content analysis to evaluate the quality of the content in order to reward it;
6) *Time to reward* shows how much time passes from performing a rewardable activity to receive the reward.

The table shows that numerous platforms consider content creation and curation activities to be rewarded. Minds is the only platform that considers rewardable a wide range of social actions, including getting one's content shared by others or being followed. Instead, DeSo adopts a unique strategy, where rewards are connected to speculating that a content creator will be popular in the future, and will be able to monetize their content. The most important peculiarity to consider is the fact that *none* of the considered reward systems includes any quality characteristics related to the content generated to compute the reward to be assigned to each piece of content. This leads to potential low-quality or harmful content issues in these platforms. For instance, videos on Odysee,[11] or posts on Steemit,[12] could contain *fake news* and still receive rewards. Therefore, in the next section, we analyze how the problem of content quality in social media has been addressed by taking into account both centralized and decentralized applications, and how it can relate to that of content moderation.

C. CONTENT QUALITY ANALYSIS AND MODERATION IN SOCIAL MEDIA

Over the years, the concept of *content quality* has been studied from different perspectives and has been identified by different concepts. These range from those associated with *structured data*, such as accuracy, completeness, consistency, timeliness, validity, and uniqueness [22], to those that have emerged more recently with the spread of large amounts of content as *semi-structured* and *unstructured data* precisely because of the Web and the Social Web.

Since UGC deeply varies in quality, this "raises the challenge of maximising its utility for a variety of end-users, particularly in the age of misinformation" [23]. To tackle this issue, most of the solutions for analyzing UGC quality

[10] https://odysee.com/

[11] https://odysee.com/@Qwinten:b/The-Trump-Assassination-is-Clearly-Staged!:7

[12] https://steemit.com/blog/@steemknowledge/did-you-know-50-the-truth-behind-organism-46-b

TABLE 1. A comparison of the most important blockchain-based reward systems.

Reward System	Content curation	Content creation	Other social activities	Other monetization	Content quality	Time to reward
Steem	✓	✓	✗	✗	✗	7 days
Hive	✓	✓	✗	✗	✗	7 days
Minds	✓	✓	✓	✗	✗	1 day
Odysee	✓	✓	✓	✓	✗	7 days
Yup	✓	✓	✗	✗	✗	1 day
RCP	✗	✓	✗	✗	✗	Up to four weeks
DeSo	✗	✗	✗	✓	✗	Instant
Pixie	✓	✓	✓	✓	✗	Instant

in OSNEMs have relied, in distinct domains, on the identification/extraction of *quality features* from the content itself, in association with (mostly) supervised approaches based on classification according to content quality categories [24], [25], [26], [27], [28], [29], [30], [31], [32]. The majority of such solutions are based on content analyses that are usually carried out just *retrospectively* and can exploit the centralized nature of OSNEMs. Each centralized platform also can make its own decisions to ensure that the quality of posted content meets a platform's guidelines, including removing harmful or inappropriate material. However, this can also leading to possible censorship issues.[13]

Decentralized Social Media, both DOSNs and BOSMs, given their distributed nature, need to develop different strategies as a gatekeeper for content quality, including *content moderation* [5]. In the latest years, several works have addressed this important research topic. The Pleroma platform has been widely studied in terms of content moderation [33]. Indeed, in [5], the authors evaluate the impact of decentralized content moderation on users in this social platform, observing that there are often *collateral damages* related to the implementation of moderation policies imposed by administrators. Another important study investigates whether and to what extent toxic content spreads in Pleroma, [34] and how to tackle this issue by proposing a system for *collaborative moderation*. Federated DOSNs can apply some policies for content moderation thanks to the usage of servers. Among such platforms, Mastodon is probably the most studied one, thanks to the possibility of retrieving social content. In [35], in particular, the decentralised content moderation strategy employed by Mastodon is analyzed, where moderation is performed by specific users, and it is always applied locally, on the server instance. Specifically, so-called *blocklisting servers* prevent users of an instance from interacting with the content of servers spreading harmful or inappropriate content.

BOSMs have the possibility of employing both "traditional" decentralized content moderation strategies and *reward strategies* as a proxy to incentivize valuable content. Referring to the first approach, since BOSMs like Steemit are based on social blockchains where content is publicly available, they can simply operate on hiding certain content on their front end. In Hive, as previously introduced, born from a fork of the Steem blockchain, many members of the community have evolved certain mechanisms and projects for moderating content, even if content removal is not possible. Minds applies a *jury system* instead of a content moderation team to check *appeals* for reported content. The jury system aims to involve users and give them a voice in deciding on the appropriateness of the content. Every time a post is moderated, the original author of the post can appeal as to why the decision should be changed. The appeal is sent to twelve unique active users who are not subscribed to the reported channel. If 75% of the jury votes in favor of the appeal, then the initial decision is overturned. Otherwise, the initial decision is upheld. The jury can not ban or remove content, it is about building a democratic tagging system. In DeSo, every node that runs on top of the DeSo blockchain, including dApps, can expose whatever subset of the posts that they want, without showing illegal or harmful content. That content would still technically be on the blockchain, but not practically accessible.

In this article, however, we focus on the second approach, i.e., the fact that BOSMs can incentivize quality content based on reward systems. Indeed, as previously introduced, rewarding systems have been proposed also to solve this issue [3]. However, their potential is not fully exploited because the current reward strategies do not take into account in any way quality characteristics of the published content and related semantic aspects when assigning rewards. Only in a recent work [36] the authors propose a study concerning *crypto-reward fairness* on user engagement. In this paper, the authors analyze the impact of fairness in the distribution of cryptocurrency rewards on user content creation in Steemit focusing on the introduction of the Hardfork19 (HF19) in June 2017. However, also in this paper, the analysis of content quality is not taken into consideration when awarding a reward.

III. EVALUATING THE EFFECTIVENESS OF BOSM REWARD SYSTEM BASED ON CONTENT ANALYSIS

A first, important part of this study is devoted to analyze the quality of content with respect to its being reward-worthy; this concerns the identification and analysis of *quality features* that can be taken into consideration for this purpose. Hence, the main *feature categories* that have been

[13] https://www.theguardian.com/media/2021/may/26/pro-palestine-censorship-facebook-instagram

used in the literature for content quality analysis in social media are introduced in this section.

Subsequently, they are employed in association with different *classifiers* to identify quality UGC, and to assess the effectiveness of reward systems in BOSMs concerning this aspect. In particular, we consider, within each feature category, those features that can be effectively extracted from the *Steemit* platform, which is used in this article to both model and evaluate the considered solutions.

A. FEATURE ENGINEERING

The six considered feature categories include the so-called: (*i*) *stylistic features*, (*ii*) *readability features*, (*iii*) *grammar features*, (*iv*) *opinion-based features*, (*v*) *metadata features*, and (*vi*) *textual representation features*.

1) STYLISTIC FEATURES

They are characteristics that are related to the identification of *patterns*, *trends*, and/or *statistics* in texts [37]. They have been considered in several solutions for UGC quality analysis in OSNEMs, such as [25], [38], [39], and [40]. They include text length, the presence of specific characters, the number of words, syllables, sentences, etc. For this work, we considered the following features.

1) **length:** the length in characters of the text of a post;
2) **words:** the number of words in the text of a post;
3) **sentences:** the number of sentences in the text of a post;
4) **syllables_count:** the number of syllables in the text of a post;
5) **word_sentence:** the number of words over the number of sentences;
6) **syllables_word:** the number of syllables over the number of words;
7) **characters_word:** the number of characters over the number of words;
8) **unique_word:** the number of unique words;
9) **n_spaces:** the number of spaces in the text;
10) **n_consonants:** the number of consonants in the text;
11) **n_numbers:** is equal to the count of the numbers in the text;
12) **contains_?:** is equal to 1 if the text contains ?, instead, is 0;
13) **contains_!:** is equal to 1 if the text contains !, instead, is 0;
14) **multiple_!:** is equal to 1 if the text contains multiple !, instead, is 0;
15) **multiple_?:** is equal to 1 if the text contains multiple ?, instead, is 0;
16) **multiple_...:** is equal to 1 if the text contains a sequence of more than three ., instead, is 0;
17) **spaces_length:** the number of spaces over the length of the text in characters;
18) **avg_l_words:** the average length of words;
19) **avg_l_sentences:** the average length of sentences;
20) **n_punctuation:** number of punctuation characters;
21) **punctuation_length:** the number of punctuation characters over the length of the text in characters;
22) **correct_capitalization:** if the text is correctly formatted.

2) READABILITY FEATURES

They measure the complexity of the written language and the ease of understanding for the intended audience. They can be computed by using factors such as word length and complexity, sentence structure, and syllables, and by means of suitable *readability indexes* [25], [41]. In fact, distinct readability indexes have been proposed and employed in the literature for UGC quality assessment in OSNEMs, such as [25] and [38]. In this work, we employed two of the most popular ones, i.e., (*i*) the *Gunning fog index* [42], and (*ii*) the *Flesch Reading Ease formula* [41]. The two readability features that were obtained are as follows:

23) **gunning_fog**: the Gunning fog index estimates the years of formal education a person needs to understand an English text on the first reading. The equation to compute this index is defined as:

$$grade_level = 0.4(\mathrm{ASL} + \mathrm{PHW})$$

where ASL indicates the *Average Sentence Length*, i.e., the number of words divided by the number of sentences, and PHW indicates the *Percentage of Hard Words*. Hard words are all the words of three or more syllables that are not proper nouns, combinations of easy words or hyphenated words, or two-syllable verbs made into three with -es and -ed endings;

24) **flesch**: the *Flesch Reading Ease formula* assesses the difficulty of reading passages written in English. It is calculated using the average number of syllables per word and the average number of words per sentence in a text. The formula produces a value ranging from 0 to 100, where a high value means that the text is easy to read. It is formally defined as:

$$\mathrm{RE} = 206.835 - (1.015 \cdot \mathrm{ASL}) - (84.6 \cdot \mathrm{ASW})$$

where RE stands for *Readability Ease*, ASL indicates the *Average Sentence Length*, i.e., the number of words divided by the number of sentences; ASW indicates the *Average number of Syllables per Word*, i.e., the number of syllables divided by the number of words. Scores between 90.0 and 100.0 indicate an easily understandable text by an average 4th grader. Scores between 60.0 and 70.0 easily understandable text by 7th and 8th graders. Scores between 0.0 and 30.0 an easily understandable text by college graduates.

3) GRAMMAR FEATURES

They capture grammatical aspects of texts that can be interpreted as quality indicators. In fact, by counting the number of adverbs, pronouns, nouns, verbs, adjectives, and other *Parts of Speech* (PoS) in the text, we can gain insights

into the overall structure and coherence of the writing. The use of such features has proven to be a powerful tool for identifying the quality of UGC in text mining applications [43], [44]. We consider the following grammar features.

25) **n_adverbs:** the number of adverbs in the text;
26) **n_verbs:** the number of verbs in the text;
27) **n_adjectives:** the number of adjectives in the text;
28) **n_pronouns:** the number of pronouns in the text;
29) **n_nouns:** the number of nouns in the text;
30) **adjectives_adverbs:** the number of adjectives plus the number of adverbs.

4) OPINION-BASED FEATURES

They are able to characterize textual content based on the author's stance and the sentiment transmitted. We considered, in particular, features related to: (*i*) *text polarity* by means of *sentiment analysis*, which gives us insight into how positive, neutral, or negative a text is; (*ii*) *text emotion* by means of *emotion recognition,* for the identification of specific emotions from a text; and (*iii*) *offensive language* by means of *offensive language identification*, which gives information on whether a text contains offensive content or not. The usage of opinion-based features has been already tested in the literature for evaluating the quality of UGC in OSNEMs, such as in [38] and [43]. In this article, we used *contextual models* to build the considered features [45], which are further detailed later in the scenario of textual representation features.

a: SENTIMENT ANALYSIS FEATURES

For the sentiment analysis task, we used a semantic-based approach, i.e., a *roBERTa-based model* [46], trained on ∼124M tweets from January 2018 to December 2021 [47], and fine-tuned for sentiment analysis with the *TweetEval benchmark* [48], constituted of social media content. Additionally, to implement this model for sentiment analysis tasks, we leveraged the Python *Transformers* library.[14] The goal of this task is to extract the sentiment polarity of a text, which can be classified into *positive*, *neutral*, and *negative*. For this reason, we built the following three features.

31) **positive**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the sentiment of the text is positive;
32) **neutral**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the sentiment of the text is neutral;
33) **negative**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the sentiment of the text is negative.

b: EMOTION RECOGNITION FEATURES

Also for the emotion recognition task, we used the roBERTa-based model from the *Transformers* library, trained on ∼58M tweets, and fine-tuned for emotion recognition with the TweetEval benchmark. This task consists of recognizing the emotion evoked by a tweet. This model of emotion recognition assigns to posts the most common four emotions among *anger*, *joy*, *sadness*, and *optimism*. Hence, we generated the following four features.

34) **anger**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the text expresses anger;
35) **joy**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the text expresses joy;
36) **sadness**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the text expresses sadness;
37) **optimism**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the text expresses optimism.

c: OFFENSIVE LANGUAGE IDENTIFICATION FEATURES

Also for this task, we used the roBERTa-based model, trained on ∼58M tweets, and fine-tuned with the TweetEval benchmark. This task consists of identifying whether some form of offensive language is present in a piece of text. In this work, we focus just on the presence or the absence of offensive language, therefore we extract only two features.

38) **offensive**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the text is offensive;
39) **not_offensive**: it is a number in a range between 0 and 1. If it is 1 or ∼ 1, the text is not offensive.

5) METADATA FEATURES

They are related to data and metadata that can be present in, or related to, a text. They allow capturing additional information regarding the content or their authors. In the literature, these features have been used for the quality analysis of UGC in OSNEMs [40], [49]. For this category, we extracted features mostly related to the content, i.e., the number of images and links included in the different posts, as well as, the category to which the posts belong, as follows.

40) **n_img**: the number of images in a post;
41) **n_link**: the number of links in a post;
42) **category**: often, posts in BOSM are categorized according to the topic to which the contents belong. In this case, instead of a single feature, we generated 33 dummy features representing the most frequently used categories in BOSMs (e.g., **photography**, **life**, **travel**, **music**, etc.). Such features will be illustrated in detail in Section IV-B2.

6) TEXTUAL REPRESENTATION FEATURES

They are related to the numeric formal representation of the semi-structured/unstructured text constituting the social media content, to make it mathematically computable. This feature category has also been used in the literature to analyze UGC [50], [51]. We describe how we built two categories of textual representation features, using the currently most advanced textual representation models, i.e., contextualized word embedding models such as BERT [52], ERNIE [53], and MISTRAL [54], which allow semantic and contextual aspects of the text to be taken into account [45]. Hence, such categories are denoted as: (*i*) *BERT-based representation*

[14]https://huggingface.co/docs/transformers/index

features, (*ii*) *ERNIE-based representation features*, and (*iii*) *MISTRAL-based representation features*.

a: BERT-BASED REPRESENTATION FEATURES

Bidirectional Encoder Representations from Transformers (BERT), already used for the extraction of opinion-based features (i.e., via the roBERTa-based model), is a *Transformer-based language model* developed by Google, with a variable number of encoder layers and self-attention heads. In our framework, given an input text, BERT is employed to produce a dense vector representation of the input that considers semantic similarity among words. Specifically, the model is a pre-trained *DistilroBERTa-based model* [55], which is fine-tuned over a 1 billion sentence pairs dataset. The model maps each text to a 768-dimensional dense feature vector that can be used for tasks like classification or semantic search.

b: ERNIE-BASED REPRESENTATION FEATURES

ERNIE 2.0 is a continual pre-training framework proposed by Baidu in 2019, which builds and learns incrementally pre-training tasks through constant multi-task learning that can be used to learn lexical, syntactic, and semantic information from several different tasks [56]. Like the BERT-based representation model employed for extracting text representation features in the above-mentioned case, the ERNIE-base model maps texts to a 768-dimensional dense vector space. So, also in this case, we produce a 768-dimensional dense feature vector for the text of each post.

c: MISTRAL-BASED REPRESENTATION FEATURES

The model used is *e5-mistral-7b-instruct* [57] initialized from *Mistral-7B-v0.1* a Large Language Model with 7 billion parameters [54]. Specifically, this model was fine-tuned on synthetic data generated by LLMs and a mixture of multilingual datasets to generate text embedding [58]. This MISTRAL-based model maps texts to a 4,096-dimensional dense vector space. So, for the text of each post, we produce a 4,096-dimensional dense feature vector.

B. QUALITY-BASED CLASSIFICATION OF CONTENT

This second phase constitutes, along with the feature engineering phase, the core of the proposed analysis. In fact, it is at this stage that distinct *classifiers* are employed, in association with the aforementioned *quality features*, to evaluate whether the content produced by users on BOSM platforms is truly worthy of reward.

The strategy considered is therefore about performing classification tasks that can assign posts generated within BOSMs to distinct categories of reward-deservingness. In general, with any supervised model, we try to learn a function that maps an input to an output based on example input-output pairs. In the solution proposed in this work, we exploit a subset of the available labeled data (the labeling process is described in Section IV-A1) as *training set* to train five different classifiers, based on well-known supervised machine learning techniques, i.e., *Support Vector Machines*, *Random Forests*, *Logistic Regression*, *Multilayer Perceptron*, and *XGBoost*. The *test set* consists of the remaining part of the available labeled dataset. The pipeline of such a strategy is illustrated in Figure 1.

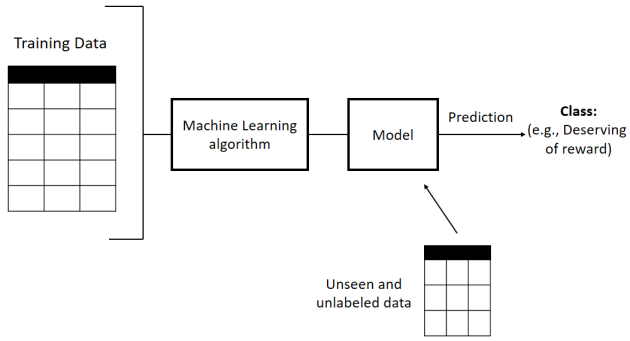

FIGURE 1. The supervised learning strategy to identify distinct classes of reward-deservingness to be associated with BOSM posts.

IV. EXPERIMENTAL EVALUATION

This section is devoted to illustrate distinct dimensions related to the performed experimental evaluation. The dataset used, the process of labeling this dataset, and the results obtained are, in particular, described.

A. THE DATASET

The dataset here described has been gathered specifically for this work. In particular, we considered the content generated by users on the *Steemit* platform, using its official API.[15] The main reason behind this choice is that Steemit stores data on the blockchain Steem using special transactions provided by the blockchain, as described in [59]. On top of that, Steemit also provides the actual rewards obtained. The *transaction* of interest for this work is the *comment*. In the Steem blockchain, it is related to the creation of both *posts* and *comments*. Moreover, this type of transaction contains all the attributes we need to build all the features described in Section III-A. So, by filtering only the posts (excluding comments) in the data coming from the blockchain, we could collect all the attributes we needed for the *feature engineering* phase.

In particular, we downloaded over 4 million Steemit posts, belonging to the January 2018–February 2018 time period. Several factors influenced this selection. This period represents the two months with the highest platform activity, as discussed in [60]. The value of STEEM, the platform's cryptocurrency, peaked in January 2018, significantly increasing the value of rewards and attracting more users motivated by these incentives. Hence, the posts from this time frame are particularly diverse in nature, covering a broad spectrum of content and topics. This may explain why many recent studies on Steemit tend to use datasets from similar or

[15] https://developers.steem.io/

even earlier time frames, often tied to specific events, such as those analyzed in [36]. This alignment with the current literature was another key reason for selecting this period as our reference point in constructing the dataset.

For each post, we collected the following post attributes:

a. **date:** the date when the post was created;
b. **author:** the author of the post;
c. **permlink:** the unique string identifier for the post, linked to the author of the post;
d. **URL:** the hyperlink of the post within Steemit platform created by concatenating the string "https://steemit.com/" + "@" + author + "/" + permlink;
e. **body:** the text of the post;
f. **parent_permlink:** it coincides with the first tag entered by the author and represents the category to which the post belongs;
g. **active_voters:** all data referring to users who voted for the post;
h. **author_reputation:** the post author's reputation;
i. **total_payout_value:** the author's rewards;
j. **curator_payout_value:** the curator's rewards;
k. **tags:** information contained in *json_metadata* parameter representing all the tags inserted by the author in a specific post;
l. **image:** information contained in *json_metadata* parameter representing all the images inserted by the author in a specific post;
m. **links:** information contained in *json_metadata* parameter representing all the links inserted by the author in a specific post.

The collected data were then subjected to a *pre-processing* phase, both to carry out data cleaning operations and to select a subset of these data that are used for the definition of the supervised models and for experimental evaluations. In particular, we performed *data cleaning* to delete all the irrelevant elements of the text of the collected posts (for example all the HTML tags, filenames, links, or email addresses), and to convert emojis to texts (e.g., :-) becomes "smiling face", :-(becomes "frowning face", etc.).

In addition, we performed a *sub-selection* of the considered data to deal with a manageable amount of them to be labeled by human assessors. To make this selection, while still keeping the data representative of the entire collection, we performed the following steps: (*i*) from the dataset we filtered posts by language, by considering only posts with text written in English, in addition, we also included all the posts that present the text in English and the respective translation into another language (which was deleted during the subsequent analysis stages to work with English content only) to collect more types of different content; (*ii*) on this English-filtered dataset, we randomly sampled 6,000 posts with no replacement (i.e., elements cannot be selected more than once), and checked that the sample maintains the same variables' distributions of the population. Once the pre-processing phase was completed, we moved on to the data labeling phase.

1) DATA LABELING BY HUMAN ASSESSORS

The *data labeling* operation performed on the sub-selection of data, is essential for both the quality-based classification of the content phase (as illustrated before, such data are used for the development of the supervised classifiers) and the testing phase. The labels, as will be detailed later, were provided by human assessors with respect to the merit of posts to receive rewards or not. The first step of the labeling phase on the sub-dataset constituted by 6,000 posts was to assign six distinct sets of 1,000 posts to six groups of three different assessors each (18 assessors in total, well versed in social media, ranging in age from 20 to 60 and gender-balanced). In this way, each post has three distinct labels. With respect to the labels given by the various assessors for each post, we calculated the *Fleiss' kappa* score [61]. Fleiss' kappa is a statistical measure for assessing the reliability of agreement between a fixed number of raters when assigning categorical ratings to a number of items or classifying items. In our case, we obtained a score $\kappa = 0.30$, which indicates *fair agreement* among assessors [62].

Having such kind of agreement, the idea to have three assessor labels for every 1,000 posts allowed us to use a *majority criterion* in defining the final label to be associated with the post. It was decided to implement this logic to reduce the intrinsic subjectivity related to human assessment. In the end, with three different assessments for each post, we obtained a total number of 18,000 assessments. Each post was labeled against *four labels* by analyzing only the content (i.e., the title and the body of the post), excluding comments. Such labels together with the associated semantics that we provided to human assessors are the following.

- *Deserving of reward:* the content of the post is original or highly informative, enriches the knowledge of those who read it or otherwise contains very useful information for the end reader. It undoubtedly deserves to be rewarded;
- *Partially deserving of reward:* the content of the post is ordinary, not fully informative, or partly incomplete, and contains interesting but easily available information that may not be very original. It partially deserves to be rewarded;
- *Not deserving of reward:* the content of the post contains meaningless phrases or content that is totally irrelevant to the end reader or is offensive and/or inappropriate, or it is just spam. Not deserving of a reward;
- *Unable to evaluate:* the assessor is unable to label the post for several reasons (e.g., the post is without text, or contains just an image or a link or a video that the assessor is not able to judge). NOTE: the presence in the post simply of an image or video does not automatically make it belong to this label. It can belong to the previous categories if the assessor is able to assess the quality of such content.

After the labeling activity carried out by the assessors, as previously illustrated, we had three different labels associated with each post. For obtaining a single label, we have applied the majority criterion previously introduced, and illustrated in Table 2. In the table, 'D' indicates the *Deserving of reward* label, 'P' the *Partially deserving of reward* label, 'N' the *Not deserving of reward* label, and 'U' the *Unable to evaluate* label. The 'F. Label' column indicates the final label associated with each post depending on the three labels assigned by three human assessors. The final label distributions over the 6,000 posts are illustrated in Table 3.

TABLE 2. Data labeling majority criterion.

Labels	Count	F. Label	Labels	Count	F. Label
NNN	1,131	N	DNP	204	P
NNP	681	N	DDN	192	D
DPP	485	P	PPU	177	P
DDP	483	D	DDU	172	D
NPP	463	P	DNU	112	P
NNU	439	N	DPU	48	P
DDD	416	D	UUU	44	U
PPP	373	P	NUU	14	N
NPU	296	N	DUU	3	U
DNN	265	N	PUU	2	U

TABLE 3. Label values distribution.

Label	Count
Deserving of reward	1,263
Partially deserving of reward	1,862
Not deserving of reward	2,826
Unable to evaluate	49

We can observe, from the obtained distribution, how the category *Unable to evaluate* is significantly under-represented, while how the categories *Partially deserving of reward* and *Deserving of reward*, taken together, have a similar numerosity compared to the category *Not deserving of reward*. These observations will be considered in Section IV-B3 in order to produce and discuss classification results.

Looking further into the characteristics that distinguish the posts belonging to the various categories and observing Figure 2, we can state that within the *Not deserving of reward* category there are posts with few images and a very short length of text in characters. Whereas, within the category *Deserving of reward* we can find posts with a lot of pictures and a much longer length in characters than those that are considered as not worthy.

Finally, analyzing the most frequently used categories within the *Deserving of reward* category, we found the topics *life* and *travel*. For the *Partially deserving of reward* category, we found the topics *photography* and *life*. For the *Not deserving of reward* category, we found the topics *photography* and *introduce yourself*.

B. RESULTS

In this section, we provide the results obtained by the assessment of different configurations of the supervised learning strategy by considering distinct classifiers based on *Support Vector Machines*, *Random Forests*, *Logistic Regression*, *Multilayer Perceptron*, and *XGBoost*, in association with distinct feature set configurations, among those considered in this work. In particular, Section IV-B1 illustrates some technical details related to the training of the classifiers; Section IV-B2 presents the considered feature sets configurations; Section IV-B3 presents and discusses the results of the distinct classifiers applied to distinct classification tasks, feature set configurations, and evaluation metrics; finally, Section IV-B4 illustrates an example of how assigning rewards to actual quality content, BOSM platforms would also benefit from substantial savings.

1) TECHNICAL DETAILS

To train the five classifiers considered, we performed *cross-validation* [63]. Specifically, we used *random permutation cross-validation* (by means of the `ShuffleSplit` model in `scikit-learn`),[16] where data are first mixed and then split into a pair of training and test sets, and a defined number of independent splits of the training/test set (in this case, 5 splits) are generated. By using this method we avoid the results depending largely on the choice of subdivision in training/test as discussed in [63]. Before training the classifiers, we performed standardization of the dataset using the `StandardScaler` model again provided by `scikit-learn`,[17] because it constitutes a common requirement for many machine learning estimators implemented in this Python library [64]. To assess the robustness of the considered machine learning models, we calculated the 95% confidence intervals for standard measures for computing classification effectiveness (i.e., *accuracy* and F1-*score*).

2) FEATURE SET CONFIGURATIONS

Several feature set configurations were tested against the above-mentioned classifiers. They are detailed below and graphically illustrated in Figure 3. Specifically, they are constituted by the *feature vectors* detailed below.

1) *Feature vector:* the 74-dimensional vector made up of all the features described in Section III-A, i.e., 41 + 33 dummy features for *category* (they are summarized in Table 4);
2) *BERT-based representation vector:* the 768-dimensional dense vector obtained by using the BERT text representation model;
3) *ERNIE-based representation vector:* the 768-dimensional dense vector obtained by using the ERNIE text representation model;

[16] https://scikit-learn.org/stable/modules/cross_validation.html#shufflesplit
[17] https://scikit-learn.org/stable/modules/preprocessing.html#preprocessing-scaler

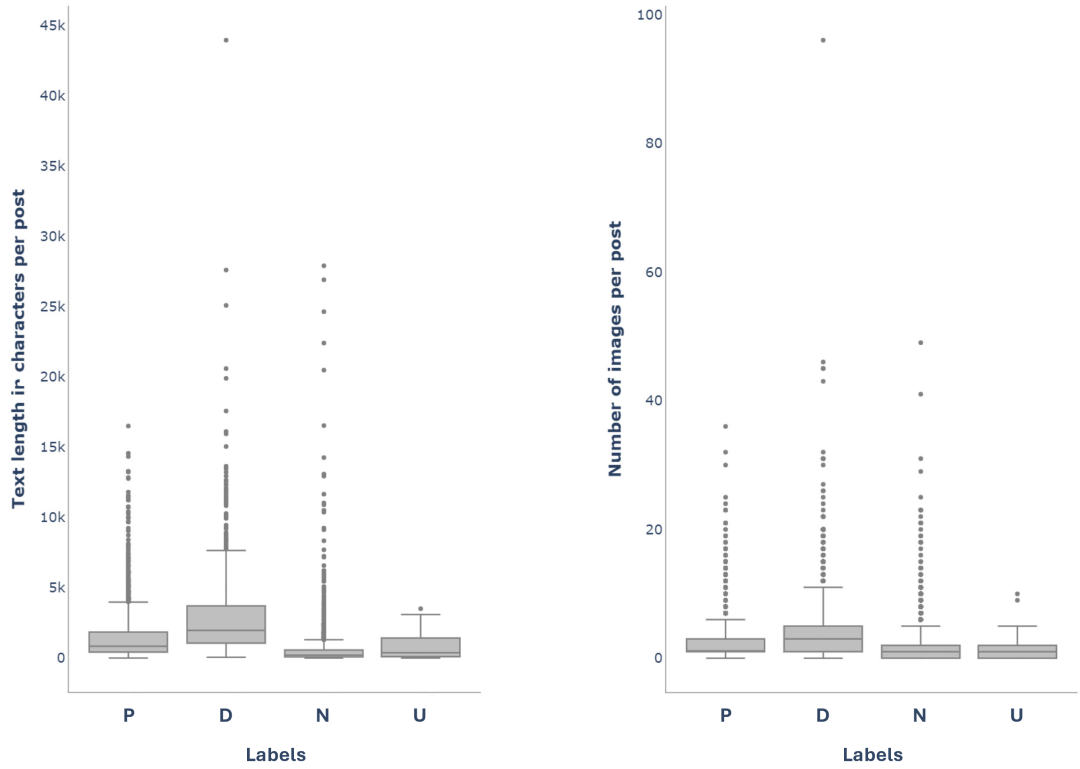

FIGURE 2. Characteristics of posts by label.

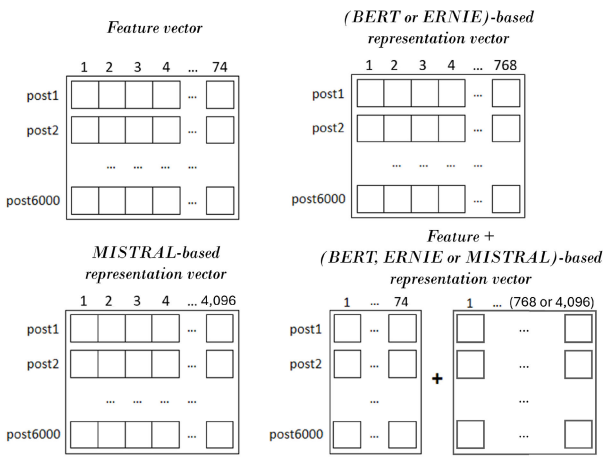

FIGURE 3. Different post representations based on distinct configurations of the considered features.

4) *MISTRAL-based representation vector:* the 4,096-dimensional dense vector obtained by using the MISTRAL text representation model;
5) *Feature + BERT-based representation vector:* the 74-dimensional feature vector + the 768-dimensional BERT-based representation vector;
6) *Feature + ERNIE-based representation vector:* the 74-dimensional feature vector + the 768-dimensional ERNIE-based representation vector.
7) *Feature + MISTRAL-based representation vector:* the 74-dimensional feature vector + the 4,096-dimensional MISTRAL-based representation vector.

3) CLASSIFICATION RESULTS

Before presenting the various results, we need to clarify that they have been obtained according to choices that were made with respect to the considered scenario.

- First of all, we would like to emphasize that our study represents a pioneering effort in comprehensively examining the interplay between content quality and rewards garnered via the BOSM reward systems. Consequently, given the absence of existing literature benchmarks addressing this specific relationship, we regrettably cannot engage in comparative evaluations against established baselines. Therefore, we have opted to utilize literature features employed in quality content classification tasks within OSNEMs, in association with various distinct baseline classifiers, for comparison;
- As delineated in Section IV-A1, it was noted that the class labeled as *Unable to evaluate* had an under-representative number of posts, rendering it negligible for inclusion in the training and testing datasets utilized by the classifiers employed. Hence, these posts and their corresponding class were excluded from consideration;

TABLE 4. Feature set constituting the feature vector: 41 + 33 dummy features for *category*.

Features		
1. length	26. n_verbs	50. *dlive*
2. word	27. n_adjectives	51. *dsound*
3. sentence	28. n_pronouns	52. *dtube*
4. syllables_count	29. n_nouns	53. *esteem*
5. word_sentence	30. adjectives_adverbs	54. *food*
6. syllables_word	31. positive	55. *funny*
7. characters_word	32. negative	56. *gaming*
8. unique_word	33. neutral	57. *health*
9. n_spaces	34. joy	58. *indonesia*
10. n_consonants	35. sadness	59. *introduceyourself*
11. n_numbers	36. optimism	60. *life*
12. contains_?	37. anger	61. *love*
13. contains_!	38. offensive	62. *money*
14. multiple_!	39. not_offensive	63. *music*
15. multiple_?	40. n_img	64. *nature*
16. multiple_...	41. n_link	65. *news*
17. spaces_length	———	66. *photography*
18. avg_l_words	42. *art*	67. *poetry*
19. avg_l_sentences	43. *bitcoin*	68. *science*
20. n_punctuation	44. *blog*	69. *steem*
21. punct_length	45. *busy*	70. *steemit*
22. correct_capitaliz	46. *christianity*	71. *story*
23. gunning_fog	47. *colorchallenge*	72. *travel*
24. flesch	48. *contest*	73. *utopian-io*
25. n_adverbs	49. *crypto*	74. *zappl*

- Moreover, based on preliminary findings, considerations of class distribution, and the challenging interpretability surrounding the distinction between *Deserving of reward* and *Partially deserving of reward*, particularly due to the inherent subjectivity of assessors, it was determined to pursue a *binary classification* task. This task involves distinguishing between the class *Not deserving of reward* and the combined set of the two aforementioned classes. Evaluation of this classification task will be conducted using conventional metrics such as *accuracy* and F1-*score*;
- Finally, we also explored the scenario of a *multi-class classification* task, where each of the three classes is considered separately. In this context, we opted to utilize *normalized Discounted Cumulative Gain* (nDCG) as a measure of effectiveness. Unlike other metrics, nDCG enables the evaluation of rankings [65] and is less sensitive to misclassifications that might occur between adjacent classes. For instance, if a post labeled by human assessors as *Partially deserving of reward* is misclassified into the *Deserving of reward* class, or vice versa, it does not significantly impact the nDCG score. Consequently, we proceed to present and elaborate on the results obtained under various evaluation scenarios.

a: BINARY CLASSIFICATION, FEATURE VECTOR

The initial findings pertain to the evaluation of the performance of the *five classifiers* in conjunction with the *feature vector* concerning the *binary classification* task. Specifically, we present the accuracy and F1-score metrics achieved while assessing the effectiveness of various *feature categories* within the feature vector. This includes analyzing the impact of *stylistic* [38], [39] + *readability* [25], [40], *grammar* [43], [44], *opinion-based* [38], [43], and *metadata features* [40], [49], both individually and in conjunction with each other. These results are illustrated in Figure 4 and further detailed in Appendix , Table 7.

From these findings, it's evident that when examining the performance of individual feature categories, *stylistic+readability features* exhibit the highest performance, achieving an accuracy of 0.76 and an F1-score of 0.79 (when utilized with *XGBoost*). Following closely, *grammar features* emerge as the second-best performing category, achieving an accuracy of 0.76 and an F1-score of 0.78 (also in conjunction with *XGBoost*). On the other hand, *metadata features* yield an accuracy of 0.65 and an F1-score of 0.66 (when employed with *XGBoost*). However, *opinion-based features* demonstrate the least effectiveness in capturing content quality, attaining an accuracy of 0.57 and an F1-score of 0.64 (again, with *XGBoost*).

Upon examining the combined effectiveness of feature categories, it becomes apparent that employing the entire 74-dimensional feature vector (comprising *stylistic*, *readability*, *grammar*, *opinion-based*, and *metadata features*) leads to a modest enhancement in classification effectiveness. This is evidenced by achieving an accuracy of 0.77 and an F1-score of 0.79 when utilizing both *Random Forests* and *XGBoost*.

These observations suggest that features associated with textual writing style and grammatical accuracy play a crucial role in identifying quality content.

b: BINARY CLASSIFICATION, FEATURE VECTOR, BERT/ERNIE-BASED REPRESENTATION VECTORS

The subsequent findings pertain to assessing the effectiveness of the *five classifiers* in conjunction with solely the *BERT-based*, *ERNIE-based*, and *MISTRAL-based representation vectors* [50], [51], as well as in combination with the *feature vector*, concerning the *binary classification* task. Results in terms of both accuracy and F1-score are illustrated in Figure 5 and further detailed in Appendix , Table 8.

As evident, the classifier exhibiting superior performance overall is the one utilizing *Random Forests* in conjunction with the 74-dimensional feature vector and the 768-dimensional BERT-based representation vector, achieving an accuracy of 0.78 and an F1-score of 0.80. Following closely, the second-best result is attained with the use of *Random Forests* as the classifier, employing the feature vector alongside the MISTRAL-based representation vector, yielding an accuracy of 0.78 and an F1-score of 0.80. The third-best outcome is observed with *XGBoost* as the classifier, utilizing the feature vector alongside the ERNIE-based representation vector, achieving an accuracy of 0.78 and an F1-score of 0.79.

Upon analyzing the efficacy of solely utilizing textual representation features (including BERT-based, ERNIE-based, and MISTRAL-based representation vectors), results closely align with those obtained using the 74-dimensional feature

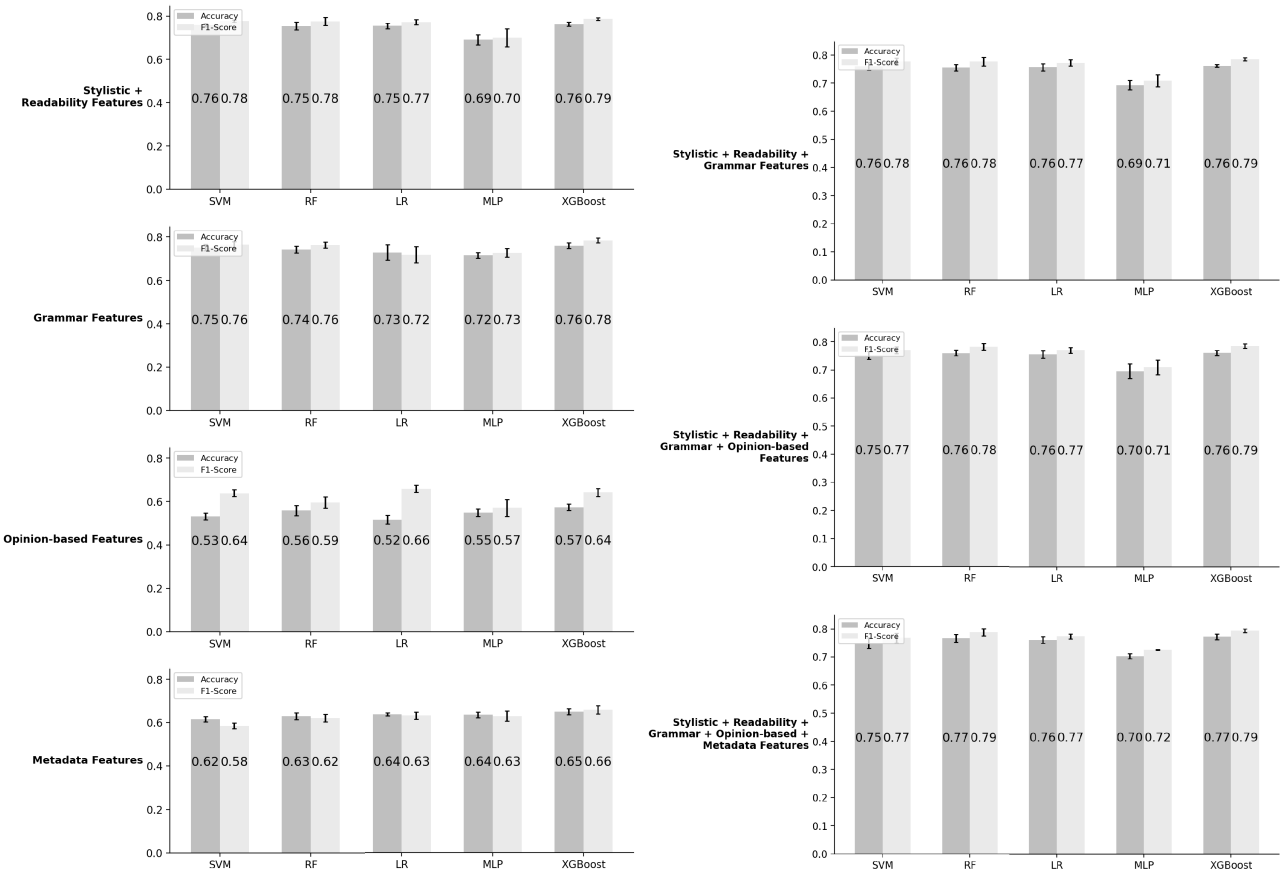

FIGURE 4. Results in terms of the effectiveness of the five classifiers in performing the binary classification task using different feature categories that are part of the feature vector alone, with a 95% confidence interval.

vector alone. Specifically, employing the MISTRAL-based representation vector yields an accuracy of 0.77 and an F1-score of 0.79 with the SVM classifier. Conversely, the BERT-based representation vector achieves an accuracy of 0.75 and an F1-score of 0.77 when paired with the SVM classifier. Similarly, employing the ERNIE-based representation vector yields comparable results with the same classifier, with an accuracy of 0.76 and an F1-score of 0.78.

TABLE 5. Results in terms of the effectiveness of the five classifiers in performing the multi-class classification task using the feature vector + the BERT-based representation vector, with a 95% confidence interval.

Model	Feature representation	nDCG
SVMs	Feature vector + BERT-based representation vector	0.82
RFs	Feature vector + BERT-based representation vector	0.84
LR	Feature vector + BERT-based representation vector	0.82
MP	Feature vector + BERT-based representation vector	0.81
XGB	Feature vector + BERT-based representation vector	0.83

c: MULTI-CLASS CLASSIFICATION, FEATURE VECTOR, BERT-BASED REPRESENTATION VECTOR

These findings pertain to the assessment, measured by nDCG, of the effectiveness of the *five classifiers* for executing the *multi-class classification* task. In this context, we solely considered the feature configuration that yielded the best overall results in the preceding evaluation, namely the *feature vector* + the *BERT-based representation vector*. The outcomes are depicted in Table 5; akin to the binary classification task, the *Random Forests* classifier emerges with the highest performance, attaining an nDCG value of 0.84 on the test dataset. This achievement is notably satisfactory, given that nDCG values are confined within the range of [0, 1]. To derive this value, in accordance with the nDCG definition [65], a ranking of posts was generated based on the classifier's predicted classes (with posts deserving of reward ranked highest, followed by partially deserving posts, and finally, those not deserving), and then compared with an ideal ranking comprised of posts labeled by human assessors.

4) REWARD ALLOCATION ANALYSIS

This section aims to analyze three primary aspects of *reward allocation* on the considered platform, based on the data collected from Steemit in the considered time period. First, it investigates how reward amounts correlate with content quality, examining whether higher-quality content receives higher rewards. Second, the analysis explores the relationship

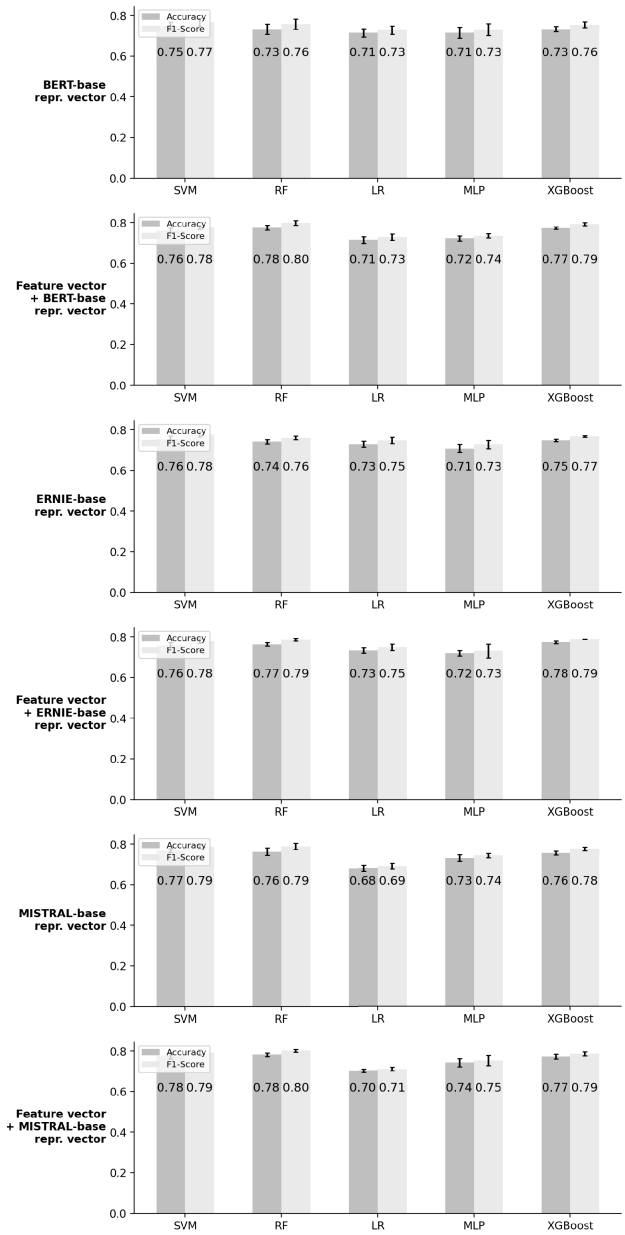

FIGURE 5. **Results in terms of the effectiveness of the five classifiers in performing the binary classification task using textual representation features alone, and in association with the feature vector, with a 95% confidence interval.**

between post engagement—measured by the number of votes—and the rewards a post garners, allowing insight into engagement-based reward patterns. Third, using Steemit as a case study, we illustrate how BOSMs could potentially reduce costs by integrating content quality aspects into their reward systems, emphasizing the benefits of more sustainable, quality-centered reward models.

a: EXAMINING REWARD MAGNITUDE W.R.T. CONTENT QUALITY

To conduct the analysis of the magnitude of rewards in the context of content quality, we examined the distribution of rewards for posts labeled as *Deserving of reward* and *Not deserving of reward*, visualizing the results through box-plots and histograms (see Fig. 6).

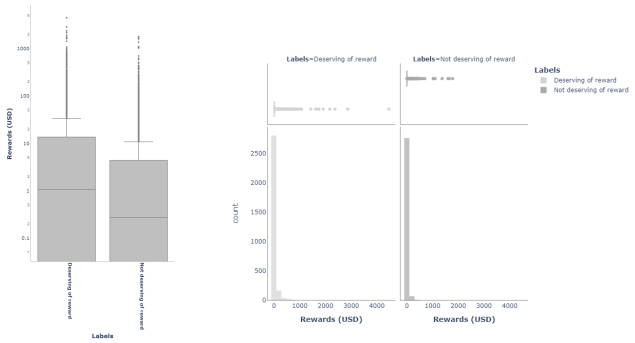

FIGURE 6. **On the left, the box-plot represents the distribution of rewards for *Deserving* and *Not deserving of reward* posts. On the right, the histogram illustrates the distribution of rewards for both categories, providing a detailed comparison between *Deserving* and *Not deserving of reward* posts.**

While a significant portion of both categories fell within similar reward brackets, typically below $15, we identified notable outliers among the *Not deserving of reward* posts, which received substantial rewards, sometimes exceeding $1,000. To further investigate *potential manipulation*, we specifically filtered posts with fewer than 10 votes that nonetheless garnered rewards of $15 or more, suspecting that such cases might indicate undue influence from powerful users, commonly referred to as "whales".

b: ANALYZING THE CORRELATION BETWEEN POST ENGAGEMENT AND REWARD DISTRIBUTION

Additionally, we examined the relationship between the number of votes a post receives and the rewards it garners. Overall, there is a moderate positive correlation between votes and rewards, reflected in an average correlation coefficient of 0.547 (see Fig. 7). This suggests that while higher votes tend to lead to greater rewards, the relationship is not strictly linear. Interestingly, for posts labeled as

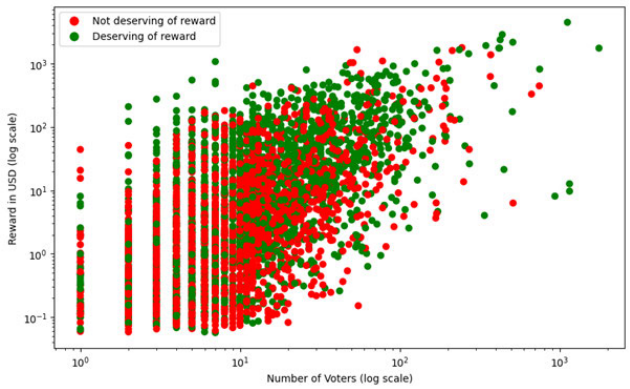

FIGURE 7. **Graph depicting the relationship between the number of votes a post receives and the corresponding reward, for posts labeled as Deserving and Not Deserving of reward.**

Not deserving of reward, the correlation coefficient drops to 0.474, indicating a weaker connection between votes and rewards. This finding supports the idea that whales—users with significant voting power—can disproportionately influence rewards for these posts, resulting in a disconnect between the number of votes and the actual rewards received.

For posts deemed *Deserving of reward*, the correlation is stronger at 0.571, implying that rewards for these posts are more closely aligned with the number of votes received. This further emphasizes the distorting effect of whale voting behavior, particularly in relation to *Not deserving of reward* content.

c: ENHANCING REWARD SYSTEMS IN BOSMS: WHAT INSIGHTS CAN WE GAIN FROM STEEMIT?

This last analysis illustrates, by using Steemit as an example, some speculations about the potential savings that similar BOSMs could achieve if their reward systems would take into account some characteristics of content quality when allocating rewards to content creators. For this purpose, we refer to the classifier that exhibited the highest performance in both binary and multi-class classifications, namely the *Random Forests* classifier, in conjunction with the most effective feature set configuration: the feature vector + the BERT-based representation vector. We identified all posts predicted by our model as *Not deserving of reward* that also received rewards from the Steemit platform, totaling approximately 600 posts out of 1,800 in the test set, and matched these predictions with assessments made by human assessors.

The potential savings for Steemit amount to 256.79 SBD (*Steem Blockchain Dollars*) in *Curators reward* and 1,220.95 SBD in *Authors reward*, totaling 1,477.74 SBD. These rewards were converted to dollars using the SBD/USD conversion rate at the payout date (7 days after post publication), resulting in $1,441.12 for *Curators reward* and $7,067.81 for *Authors reward*, as summarized in Table 6.

TABLE 6. Amount of estimated SBD and USD that Steem could have saved not rewarding the 600 posts classified as *Not deserving of reward* in the test dataset (1,800 posts), and coincident to the label provided by human assessors.

Reward	SBD	USD estimated
Curators reward	256.79	1,441.12
Authors reward	1,220.95	7,067.81

This illustrative example underscores that by considering only the 1,800 posts in the test set, the platform could have saved $8,545.55. Extrapolating this to the 4.3 million posts collected during the January-February 2018 observation period, the potential savings in cryptocurrency for Steemit become considerable. However, it is important to emphasize that the values provided in Table 6 are given just as examples to illustrate the potential of considering content quality when distributing rewards. It is essential to acknowledge the following points: (*i*) the classifier used in the example, while effective, does not achieve 100% accuracy, considering the nuanced nature of quality assessment. In our example, we only considered posts labeled as *True Positives* (TP) for calculating savings, recognizing that other posts may also warrant redistribution based on human assessor labels; (*ii*) any such solution should be viewed as a supportive tool for reward decisions and must be incorporated into concrete reward strategies implemented in real-world reward systems. Nevertheless, tackling this issue is beyond the scope of this article, and is left as future research.

V. CONCLUSION AND FUTURE WORK

Current reward systems in Blockchain-based Online Social Media (BOSMs), developed to promote the quality of content disseminated on these platforms, are largely influenced by the social interactions of users on the platform. These systems face numerous challenges, including the prevalence of bots and potential manipulation of voting activities. Such issues can inadvertently undermine the very objectives for which these platforms were designed. For instance, they may facilitate the exploitation of reward systems for undeserved gains, even in the presence of spammy content, while also contributing to the creation of echo chambers and information bubbles. Therefore, it is essential to be able to accurately identify the quality of content that is rewarded on BOSMs.

To contribute to this aim, in this article we provided a study concerning the actual relationship between the quality of the content disseminated in the context of BOSMs, and the rewards provided to users by their reward systems. We considered distinct sets of features characterizing quality content in social media, in association with distinct classifiers to identify content actually worthy of reward. To both perform and asses such a study, we exploited a real dataset, collected from Steemit, one of the most well-known decentralized social media platforms based on blockchain technology. Specifically, we hired human assessors to label the dataset with respect to reward-deservingness classes, to both train and evaluate distinct configurations of the different feature sets in association with the considered classifiers.

Based on the obtained findings, we have demonstrated that it is indeed possible to effectively assess the quality of content circulated on BOSMs by considering content quality features in association with machine learning. This capability has been utilized in this paper to distinguish between content that truly deserves rewards and that which does not. Through this analysis, we have shown that, actually, current reward systems do not fulfill their intended purpose of promoting the highest-quality content through reward strategies. Consequently, these systems fall short of functioning as de facto bottom-up content moderation mechanisms.

A challenging research direction could involve proposing and formalizing a reward system that incorporates content analysis characteristics to effectively distribute rewards

TABLE 7. Results of different feature vector's categories in performing binary classification in terms of *accuracy* and F1-*score* with a 95% confidence interval.

Feature Category	Classifier	Accuracy	C.I. Accuracy	F1-score	C.I. F1-score
Stylistic + Readability Features [25], [38] [39], [40]	SVM	0.758	[0.749, 0.767]	0.776	[0.770, 0.783]
	RF	0.753	[0.735, 0.771]	0.775	[0.756, 0.795]
	LR	0.754	[0.741, 0.767]	0.772	[0.760, 0.784]
	MLP	0.690	[0.667, 0.713]	0.699	[0.657, 0.741]
	XGBoost	0.762	[0.753, 0.772]	0.787	[0.781, 0.793]
Grammar Features [43], [44]	SVM	0.751	[0.736, 0.765]	0.764	[0.749, 0.780]
	RF	0.742	[0.727, 0.757]	0.763	[0.749, 0.777]
	LR	0.729	[0.694, 0.763]	0.718	[0.681, 0.755]
	MLP	0.716	[0.703, 0.729]	0.727	[0.707, 0.748]
	XGBoost	0.760	[0.747, 0.772]	0.784	[0.773, 0.796]
Opinion-based Features [38], [43]	SVM	0.531	[0.516, 0.546]	0.639	[0.623, 0.654]
	RF	0.558	[0.535, 0.581]	0.595	[0.569, 0.620]
	LR	0.516	[0.496, 0.536]	0.658	[0.641, 0.674]
	MLP	0.548	[0.531, 0.566]	0.570	[0.531, 0.609]
	XGBoost	0.573	[0.558, 0.587]	0.641	[0.623, 0.658]
Metadata Features [40], [49]	SVM	0.616	[0.604, 0.628]	0.585	[0.572, 0.598]
	RF	0.629	[0.613, 0.644]	0.621	[0.604, 0.638]
	LR	0.638	[0.631, 0.644]	0.632	[0.615, 0.648]
	MLP	0.636	[0.623, 0.648]	0.630	[0.607, 0.653]
	XGBoost	0.650	[0.637, 0.663]	0.659	[0.640, 0.677]
Stylistic + Readability + Grammar Features	SVM	0.760	[0.746, 0.774]	0.777	[0.765, 0.789]
	RF	0.755	[0.743, 0.767]	0.777	[0.762, 0.792]
	LR	0.756	[0.743, 0.770]	0.772	[0.761, 0.782]
	MLP	0.693	[0.676, 0.711]	0.709	[0.688, 0.729]
	XGBoost	0.761	[0.756, 0.767]	0.785	[0.779, 0.792]
Stylistic + Readability + Grammar + Opinion-based Features	SVM	0.752	[0.738, 0.767]	0.770	[0.758, 0.782]
	RF	0.760	[0.750, 0.770]	0.782	[0.770, 0.793]
	LR	0.755	[0.742, 0.767]	0.770	[0.760, 0.779]
	MLP	0.696	[0.670, 0.722]	0.709	[0.683, 0.735]
	XGBoost	0.761	[0.752, 0.770]	0.785	[0.777, 0.793]
Stylistic + Readability + Grammar + Opinion-based + Metadata Features	SVM	0.749	[0.730, 0.768]	0.767	[0.750, 0.784]
	RF	0.766	[0.752, 0.781]	0.788	[0.775, 0.801]
	LR	0.760	[0.749, 0.771]	0.773	[0.764, 0.783]
	MLP	0.702	[0.693, 0.712]	0.725	[0.723, 0.727]
	XGBoost	0.771	[0.761, 0.781]	0.793	[0.787, 0.799]

among creators. In this paper, we speculated on the potential benefits of integrating quality considerations into reward systems, which may enhance their overall effectiveness and economic viability. However, solutions that introduce quality metrics should be thoroughly evaluated to avoid exacerbating centralization issues, which could lead to additional open issues. For example, ensuring that modifications do not concentrate power among a small group of users is vital to uphold the platform's democratic values. Striking a balance between reward distribution and equitable access to resources could be crucial for nurturing a healthy and decentralized ecosystem.

TABLE 8. Results of textual representation features alone, and in association with the feature vector, in performing binary classification in terms of *accuracy* and F1-*score* with a 95% confidence interval.

Feature Set Configuration	Classifier	Accuracy	C.I. Accuracy	F1-score	C.I. F1-score
BERT-based representation vector [50], [51]	SVM	0.749	[0.730, 0.768]	0.768	[0.749, 0.787]
	RF	0.733	[0.708, 0.758]	0.759	[0.734, 0.784]
	LR	0.715	[0.694, 0.736]	0.729	[0.709, 0.749]
	MLP	0.715	[0.688, 0.741]	0.731	[0.702, 0.759]
	XGBoost	0.734	[0.721, 0.746]	0.755	[0.740, 0.769]
Feature vector + BERT-based representation vector	SVM	0.762	[0.751, 0.773]	0.777	[0.768, 0.787]
	RF	0.781	[0.764, 0.788]	0.800	[0.788, 0.810]
	LR	0.715	[0.698, 0.733]	0.729	[0.714, 0.744]
	MLP	0.723	[0.711, 0.736]	0.736	[0.726, 0.746]
	XGBoost	0.774	[0.768, 0.780]	0.793	[0.785, 0.800]
ERNIE-based representation vector	SVM	0.755	[0.741, 0.770]	0.778	[0.766, 0.789]
	RF	0.742	[0.731, 0.753]	0.761	[0.753, 0.770]
	LR	0.730	[0.715, 0.746]	0.749	[0.733, 0.766]
	MLP	0.709	[0.688, 0.731]	0.729	[0.709, 0.749]
	XGBoost	0.748	[0.742, 0.754]	0.768	[0.765, 0.771]
Feature vector + ERNIE-based representation vector	SVM	0.760	[0.746, 0.775]	0.780	[0.769, 0.791]
	RF	0.765	[0.756, 0.774]	0.788	[0.782, 0.794]
	LR	0.734	[0.719, 0.750]	0.750	[0.734, 0.767]
	MLP	0.720	[0.706, 0.733]	0.732	[0.697, 0.766]
	XGBoost	0.775	[0.768, 0.781]	0.795	[0.790, 0.799]
MISTRAL-based representation vector	SVM	0.768	[0.759, 0.778]	0.788	[0.779, 0.798]
	RF	0.763	[0.745, 0.780]	0.789	[0.774, 0.804]
	LR	0.681	[0.666, 0.697]	0.692	[0.677, 0.706]
	MLP	0.732	[0.716, 0.747]	0.744	[0.733, 0.755]
	XGBoost	0.757	[0.747, 0.767]	0.776	[0.768, 0.784]
Feature vector + MISTRAL-based representation vector	SVM	0.778	[0.763, 0.793]	0.793	[0.779, 0.810]
	RF	0.783	[0.774, 0.791]	0.800	[0.795, 0.810]
	LR	0.703	[0.695, 0.711]	0.711	[0.702, 0.721]
	MLP	0.742	[0.721, 0.763]	0.753	[0.728, 0.777]
	XGBoost	0.772	[0.760, 0.783]	0.786	[0.776, 0.797]

RESEARCH DATA/CODE AVAILABILTY

Research data and code will be made available upon request, by making a request to the Corresponding Author.

APPENDIX

TABLES OF RESULTS

This appendix details the results of the comparative assessments that have been graphically illustrated in Section IV-B3. In particular, the results related to the effectiveness of different feature categories within the feature vector to perform binary classification and the results related to the use of textual features alone (versus different representation models) and in association with the feature vector are illustrated.

REFERENCES

[1] A. Datta, S. Buchegger, L.-H. Vu, T. Strufe, and K. Rzadca, "Decentralized online social networks," in *Handbook of Social Network Technologies and Applications*. New York, NY, USA: Springer, 2010, pp. 349–378.

[2] M. Viviani and G. Pasi, "Credibility in social media: Opinions, news, and health information—A survey," *WIREs Data Mining Knowl. Discovery*, vol. 7, no. 5, pp. e120–9, Sep. 2017.

[3] B. Guidi, "When blockchain meets online social networks," *Pervas. Mobile Comput.*, vol. 62, Feb. 2020, Art. no. 101131.

[4] A. Kazdin, *The Token Economy: A Review and Evaluation*. New York, NY, USA: Springer, 2012.

[5] A. I. Hassan, A. Raman, I. Castro, H. B. Zia, E. De Cristofaro, N. Sastry, and G. Tyson, "Exploring content moderation in the decentralised Web: The pleroma case," in *Proc. 17th Int. Conf. Emerg. Netw. Exp. Technol.*, Dec. 2021, pp. 328–335.

[6] A. Z. Rozenshtein, "Moderating the fediverse: Content moderation on distributed social media," *J. Free Speech Law*, vol. 3, p. 217, Jan. 2023.

[7] B. Guidi, A. Michienzi, and L. Ricci, "A graph-based socioeconomic analysis of steemit," *IEEE Trans. Computat. Social Syst.*, vol. 8, no. 2, pp. 365–376, Apr. 2021.

[8] B. Guidi and A. Michienzi, "How to reward the Web: The social dApp yup," *Online Social Netw. Media*, vol. 31, Sep. 2022, Art. no. 100229.

[9] B. Guidi and A. Michienzi, "Users and bots behaviour analysis in blockchain social media," in *Proc. 7th Int. Conf. Social Netw. Anal., Manage. Secur. (SNAMS)*, Dec. 2020, pp. 1–8.

[10] Y. Usui, F. Toriumi, and T. Sugawara, "Impact of monetary rewards on users' behavior in social media," in *Proc. Int. Conf. Complex Netw. Appl.*, vol. 1. Cham, Switzerland: Springer, Jan. 2022, pp. 632–643.

[11] D. Nikolov, D. F. M. Oliveira, A. Flammini, and F. Menczer, "Measuring online social bubbles," *PeerJ Comput. Sci.*, vol. 1, p. e38, Dec. 2015.

[12] G. Lefebvre, O. Deroy, and B. Bahrami, "The roots of polarization in the individual reward system," *Proc. Roy. Soc. B, Biol. Sci.*, vol. 291, no. 2017, Feb. 2024, Art. no. 20232011.

[13] R. Maull, P. Godsiff, C. Mulligan, A. Brown, and B. Kewell, "Distributed ledger technology: Applications and implications," *Strategic Change*, vol. 26, no. 5, pp. 481–489, Sep. 2017.

[14] S. Squarepants, "Bitcoin: A peer-to-peer electronic cash system," *SSRN Electron. J.*, p. 21260, 2008. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3440802

[15] M. Kolvart, M. Poola, and A. Rull, "Smart contracts," in *The Future of Law and Etechnologies*. Cham, Switzerland: Springer, 2016, pp. 133–147.

[16] D. Vujicic, D. Jagodic, and S. Randic, "Blockchain technology, bitcoin, and ethereum: A brief overview," in *Proc. 17th Int. Symp. INFOTEH-JAHORINA (INFOTEH)*, Mar. 2018, pp. 1–6.

[17] A. Ahmed, S. Abdullah, M. Bukhsh, I. Ahmad, and Z. Mushtaq, "An energy-efficient data aggregation mechanism for IoT secured by blockchain," *IEEE Access*, vol. 10, pp. 11404–11419, 2022.

[18] A. Ahmed, S. Abdullah, S. Iftikhar, I. Ahmad, S. Ajmal, and Q. Hussain, "A novel blockchain based secured and QoS aware IoT vehicular network in edge cloud computing," *IEEE Access*, vol. 10, pp. 77707–77722, 2022.

[19] M. Morsidi, S. Tajuddin, R. K. Patchmuthu, and S. H. S. Newaz, "Blockchain-based reward system: A means for providing incentive to students for teaching feedback," in *Proc. Int. Conf. Electron., Commun. Inf. Technol. (ICECIT)*, Sep. 2021, pp. 1–5.

[20] B. Guidi and A. Michienzi, "The side effect of ERC-20 standard in social media platforms," in *Proc. Int. Conf. Social Informat.* Springer, Jan. 2022, pp. 114–127.

[21] Y. Zhan, Y. Xiong, and X. Xing, "A conceptual model and case study of blockchain-enabled social media platform," *Technovation*, vol. 119, Jan. 2023, Art. no. 102610.

[22] C. Batini and M. Scannapieco, *Data and Information Quality*. Cham, Switzerland: Springer, 2016.

[23] E. Momeni, C. Cardie, and N. Diakopoulos, "How to assess and rank user-generated content on Web," in *Proc. Companion Web Conf.*, 2018, pp. 489–493.

[24] E. Afful-Dadzie, A. Afful-Dadzie, and S. B. Egala, "Social media in health communication: A literature review of information quality," *Health Inf. Manage. J.*, vol. 52, no. 1, pp. 3–17, Jan. 2023.

[25] E. Agichtein, C. Castillo, D. Donato, A. Gionis, and G. Mishne, "Finding high-quality content in social media," in *Proc. Int. Conf. Web Search Web Data Mining (WSDM)*, 2008, p. 183.

[26] E. Bassani and M. Viviani, "Automatically assessing the quality of Wikipedia contents," in *Proc. 34th ACM/SIGAPP Symp. Appl. Comput.*, Apr. 2019, pp. 804–807.

[27] F. Figueiredo, H. Pinto, F. Belém, J. Almeida, M. Gonçalves, D. Fernandes, and E. Moura, "Assessing the quality of textual features in social media," *Inf. Process. Manage.*, vol. 49, no. 1, pp. 222–247, Jan. 2013.

[28] J. Fontanarava, G. Pasi, and M. Viviani, "An ensemble method for the credibility assessment of user-generated content," in *Proc. Int. Conf. Web Intell.*, Aug. 2017, pp. 863–868.

[29] H. Paul and A. Nikolaev, "Fake review detection on online e-commerce platforms: A systematic literature review," *Data Mining Knowl. Discovery*, vol. 35, no. 5, pp. 1830–1881, Sep. 2021.

[30] C. Shao, G. L. Ciampaglia, O. Varol, K.-C. Yang, A. Flammini, and F. Menczer, "The spread of low-credibility content by social bots," *Nature Commun.*, vol. 9, no. 1, pp. 1–9, Nov. 2018.

[31] C. Zhou, K. Li, and Y. Lu, "Linguistic characteristics and the dissemination of misinformation in social media: The moderating effect of information richness," *Inf. Process. Manage.*, vol. 58, no. 6, Nov. 2021, Art. no. 102679.

[32] W. Zhuang, Q. Zeng, Y. Zhang, C. Liu, and W. Fan, "What makes user-generated content more helpful on social media platforms? Insights from creator interactivity perspective," *Inf. Process. Manage.*, vol. 60, no. 2, Mar. 2023, Art. no. 103201.

[33] I. H. Anaobi, A. Raman, I. Castro, H. B. Zia, D. Ibosiola, and G. Tyson, "Will admins cope? Decentralized moderation in the fediverse," in *Proc. ACM Web Conf.*, Apr. 2023, pp. 3109–3120.

[34] H. Bin Zia, A. Raman, I. Castro, I. Hassan Anaobi, E. De Cristofaro, N. Sastry, and G. Tyson, "Toxicity in the decentralized Web and the potential for model sharing," *Proc. ACM Meas. Anal. Comput. Syst.*, vol. 6, no. 2, pp. 1–25, 2022.

[35] C. A. Bono, L. La Cava, L. Luceri, and F. Pierri, "An exploration of decentralized moderation on mastodon," in *Proc. ACM Web Sci. Conf.*, May 2024, pp. 53–58.

[36] W. Yang, Y. Kim, T. H. Kim, C. H. Lee, and Y. Ceran, "From whales to minnows: The impact of crypto-reward fairness on user engagement in social media," *Decis. Support Syst.*, vol. 185, Oct. 2024, Art. no. 114289.

[37] R. Moura, R. Sousa-Silva, and H. L. Cardoso, "Automated fake news detection using computational forensic linguistics," in *Proc. Prog. Artif. Intell.*, G. Marreiros, F. S. Melo, N. Lau, H. L. Cardoso, and L. P. Reis, Eds., 2021, pp. 788–800.

[38] A. S. Abrahams, W. Fan, G. A. Wang, Z. Zhang, and J. Jiao, "An integrated text analytic framework for product defect discovery," *Prod. Operations Manage.*, vol. 24, no. 6, pp. 975–990, Jun. 2015.

[39] K. A. Qureshi, R. A. S. Malick, and M. Sabih, "Social media and microblogs credibility: Identification, theory driven framework, and recommendation," *IEEE Access*, vol. 9, pp. 137744–137781, 2021.

[40] J. Ito, J. Song, H. Toda, Y. Koike, and S. Oyama, "Assessment of tweet credibility with LDA features," in *Proc. 24th Int. Conf. World Wide Web*, May 2015, pp. 953–958.

[41] W. H. DuBay, *Smart Language: Readers, Readability, and the Grading of Text*, 2017.

[42] R. Gunning, "The technique of clear writing,". New York, NY, USA: McGraw-Hill, 1952.

[43] D. Markovikj, S. Gievska, M. Kosinski, and D. Stillwell, "Mining Facebook data for predictive personality modeling," in *Proc. Int. AAAI Conf. Web Social Media*, Aug. 2021, vol. 7, no. 2, pp. 23–26.

[44] V. L. Rubin and N. J. Conroy, "Challenges in automated deception detection in computer-mediated communication," *Proc. Amer. Soc. Inf. Sci. Technol.*, vol. 48, no. 1, pp. 1–4, 2011.

[45] A. Miaschi and F. Dell'Orletta, "Contextual and non-contextual word embeddings: An in-depth linguistic investigation," in *Proc. 5th Workshop Represent. Learn. NLP*, 2020, pp. 110–119.

[46] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," 2019, *arXiv:1907.11692*.

[47] D. Loureiro, F. Barbieri, L. Neves, L. Espinosa Anke, and J. Camacho-Collados, "TimeLMs: Diachronic language models from Twitter," 2022, *arXiv:2202.03829*.

[48] F. Barbieri, J. Camacho-Collados, L. Neves, and L. Espinosa-Anke, "TweetEval: Unified benchmark and comparative evaluation for tweet classification," 2020, *arXiv:2010.12421*.

[49] D. Martens and W. Maalej, "Towards understanding and detecting fake reviews in app stores," *Empirical Softw. Eng.*, vol. 24, no. 6, pp. 3316–3355, Dec. 2019.

[50] M. Mozafari, R. Farahbakhsh, and N. Crespi, "Hate speech detection and racial bias mitigation in social media based on BERT model," *PLoS ONE*, vol. 15, no. 8, Aug. 2020, Art. no. e0237861.

[51] K. Roitero, C. Bozzato, V. D. Mea, S. Mizzaro, and G. Serra, "Twitter goes to the doctor: Detecting medical tweets using machine learning and BERT," in *Proc. SIIRH@ECIR*, Jan. 2020, pp. 1–3.

[52] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[53] Y. Sun, S. Wang, Y. Li, S. Feng, X. Chen, H. Zhang, X. Tian, D. Zhu, H. Tian, and H. Wu, "ERNIE: Enhanced representation through knowledge integration," 2019, *arXiv:1904.09223*.
[54] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, and W. El Sayed, "Mistral 7B," 2023, *arXiv:2310.06825*.
[55] V. Sanh, L. Debut, J. Chaumond, and T. Wolf. (2022). *Distil-RoBERTa Base Model*. Accessed: Aug. 5, 2022. [Online]. Available: https://huggingface.co/distilroberta-base
[56] Y. Sun, S. Wang, Y. Li, S. Feng, H. Tian, H. Wu, and H. Wang, "ERNIE 2.0: A continual pre-training framework for language understanding," 2019, *arXiv:1907.12412*.
[57] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei. (2024). *E5-mistral-7b-instruct*. Accessed: Feb. 14, 2024. [Online]. Available: https://huggingface.co/intfloat/e5-mistral-7b-instruct
[58] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei, "Improving text embeddings with large language models," 2023, *arXiv:2401.00368*.
[59] B. Guidi, A. Michienzi, and L. Ricci, "Steem blockchain: Mining the inner structure of the graph," *IEEE Access*, vol. 8, pp. 210251–210266, 2020.
[60] B. Guidi, A. Michienzi, and L. Ricci, "Incremental communication patterns in blockchain online social media," in *Proc. 37th ACM/SIGAPP Symp. Appl. Comput.*, Apr. 2022, pp. 1764–1770.
[61] J. L. Fleiss, B. Levin, and M. C. Paik, *Statistical Methods for Rates and Proportions*. Hoboken, NJ, USA: Wiley, 2013.
[62] J. R. Landis and G. G. Koch, "An application of hierarchical kappa-type statistics in the assessment of majority agreement among multiple observers," *Biometrics*, vol. 33, no. 2, p. 363, Jun. 1977.
[63] P. Refaeilzadeh, L. Tang, and H. Liu, "Cross-validation," in *Proc. Encyclopedia Database Syst.*, Jan. 2009, pp. 532–538.
[64] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. J. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Jan. 2011.
[65] Y. Wang, L. Wang, Y. Li, D. He, T. Liu, and W. Chen, "A theoretical analysis of NDCG type ranking measures," in *Proc. Conf. Learn. Theory*, Jan. 2013, pp. 25–54.

DAVIDE MANCINO received the M.Sc. degree in data science from the Department of Computer Science, Systems, and Communication, Università degli Studi di Milano-Bicocca, Italy, in 2022. He is currently pursuing the joint Ph.D. degree (Italian National Ph.D. in Blockchain) with the University of Camerino, Italy, and the Department of Computer Science, Systems, and Communication, Università degli Studi di Milano-Bicocca, Italy. His research interests include blockchain and blockchain online social media, social network analysis, data science, and natural language processing.

BARBARA GUIDI received the B.Sc., M.Sc., and Ph.D. degrees in computer science from the University of Pisa, Italy, in 2007, 2011, and 2015, respectively. She is currently an Assistant Professor with the Department of Computer Science, Università degli Studi di Pisa. Her current research interests include decentralized online social networks, blockchain technology, and social network analysis. She has been involved in the TPC of several international conferences and workshops. She was the Co-Chair of the conference EAI GoodTechs 2017 and several workshops. She has been a Reviewer of relevant journals, such as IEEE Access.

ANDREA MICHIENZI received the Ph.D. degree in computer science from the Department of Computer Science, University of Pisa, Italy, in July 2021. He is currently an Assistant Professor with the Department of Computer Science, Università degli Studi di Pisa. His main research interests include the dynamics of socio-economic networks, enabling technologies for the metaverse, and the application of blockchain to social networking platforms. He is one of the organizers of the Open Challenges in Online Social Networks Workshop.

MARCO VIVIANI received the M.Sc. and Ph.D. degrees in computer science from the University of Milan-La Statale, Italy. Later, he was a Postdoctoral Fellow at both the University of Insubria, Italy, and foreign institutions (University of Burgundy, INSA Lyon, France). He is currently an Associate Professor with the Department of Informatics Systems, and Communication, Università degli Studi di Milano-Bicocca, Italy. He is involved in organizing several research initiatives at the international level. His main research interests include social computing, information retrieval, natural language processing, security, privacy, and trust. On these topics, he has published more than 90 research works in international journals, international conferences, and monographs and book chapters. He was the General Co-Chair of MDAI 2019 and organized several workshops and special tracks at international conferences. He is an Associate Editor of *Social Network Analysis and Mining* and *Frontiers in Artificial Intelligence-Natural Language Processing* and an Area Editor (Web Intelligence and E-Services) of the *International Journal of Computational Intelligence Systems*.